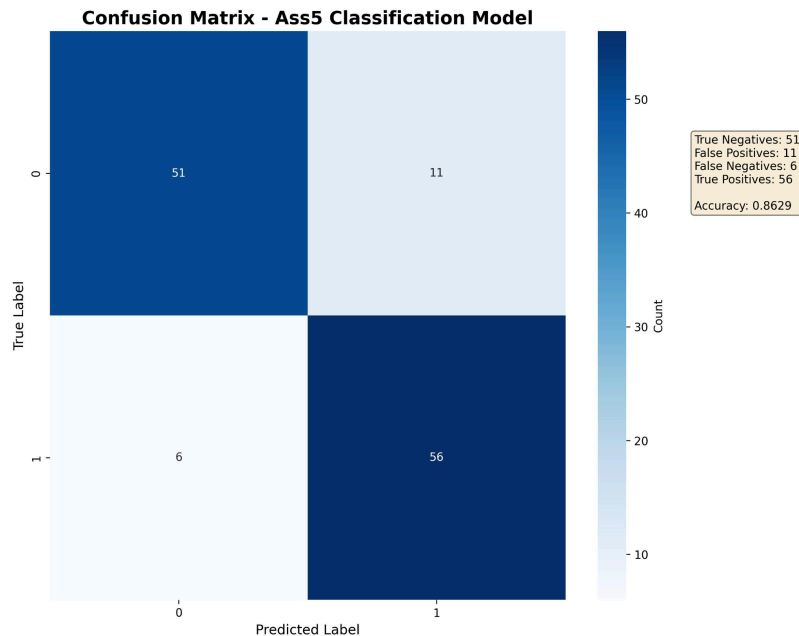


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Results:

Data Preparation

I prepared the dataset by first cleaning both the training and test files to ensure consistent formatting. I trimmed column names, standardized categorical text entries, and corrected label inconsistencies such as “Fe Male” to “Female,” “Free Lancer” to “Freelancer,” and “Unmarried” to “Single.” I kept ProdTaken as a binary target (0 = no purchase, 1 = purchase). For model development, I used ass5train.csv and split it into training and validation sets with an 80:20 stratified split, which preserved class balance (891 rows for split-training and 223 rows for validation). I used ass5test.csv only for final evaluation, so no test information leaked into model selection or threshold tuning.

Analysis

The data includes both customer profile information and interaction behavior, which are both important for predicting purchase decisions. Profile variables such as age, occupation, income, marital status, and city tier provide background context, while behavioral variables such as duration of pitch, number of follow-ups, and pitch satisfaction score reflect customer engagement and intent. From this, I observed that behavioral features are likely stronger direct signals of conversion, while demographic features help refine segmentation. I also considered class imbalance and chose to monitor recall and AUC along with accuracy, since accuracy alone may hide weak performance on the positive class.

Feature Extraction

I selected all available non-target columns as predictive features because each captures a relevant part of customer behavior or profile. Numeric features were kept numeric and missing values were imputed with the median. Categorical features were imputed with the most frequent class and then one-hot encoded so the model could learn category-specific effects without false ordinal assumptions. This preprocessing approach was applied through a single pipeline to

keep feature handling consistent between training and inference. The final feature set included variables such as Age, TypeofContact, DurationOfPitch, NumberOfFollowups, ProductPitched, PitchSatisfactionScore, MonthlyIncome, and other related attributes.

Building the Model

I built and compared three tree-based classification models inside the same preprocessing pipeline: Decision Tree, Random Forest, and Extra Trees. Since these are ensemble tree methods, the model design does not use hidden layers, neural-network optimizers, or gradient-based loss functions like binary cross-entropy. Instead, I improved robustness by using class weighting (balanced or balanced_subsample) and a large number of estimators (800 trees for Random Forest and Extra Trees). I selected the best model based on validation performance after threshold tuning on predicted probabilities. Extra Trees performed best, and I saved it as the final model with a tuned threshold of 0.485.

Evaluation Results

On the validation split, the selected model achieved strong performance with accuracy 0.9013, recall 0.9189, precision 0.8870, F1-score 0.9027, and AUC 0.9521. On the held-out test set, the model achieved accuracy 0.8629, recall 0.9032, precision 0.8358, F1-score 0.8682, and AUC 0.9242.

The confusion matrix image supports these test results directly. It shows True Negatives = 51, False Positives = 11, False Negatives = 6, and True Positives = 56, and the annotation box on the right clearly states Accuracy: 0.8629. This confirms that the model correctly classifies most positive cases while maintaining good overall accuracy.